Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100307

Subject: Electrical Machines-I

Time: 03 Hours
Full Marks: 70

Instructions: -
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

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Q.1 *Choose the correct option / answer the following (Any seven question only):* [2 x 7 = 14]

(a) Which law gives the magnetic field due to a small current element?
(i) Gauss's Law (ii) Ampere's Law
(iii) Biot–Savart Law (iv) Faraday's Law

(b) In a magnetic circuit, the reluctance of air is:
(i) Very low (ii) Same as iron
(iii) Very high (iv) Zero

(c) When a magnetic material is magnetized, the lagging of B behind H is called:
(i) Retentivity (ii) Coercivity
(iii) Eddy current (iv) Hysteresis

(d) Which device uses torque due to magnetic interaction to show current measurement?
(i) Lifting magnet (ii) Relay
(iii) Galvanometer (iv) Transformer

(e) Harmonics in magnetizing current arise due to:
(i) Linear B-H curve. (ii) Nonlinear B-H curve.
(iii) Leakage reactance (iv) Load resistance.

(f) Which loss remains constant at all loads?
(i) Eddy current loss (ii) Armature copper loss.
(iii) Stray load losses (iv) Brush contact loss

(g) Increasing the number of poles in a DC machine:
(i) Reduces flux (ii) Reduces size of armature conductors
(iii) Increases frequency (iv) Reduces armature reaction

(h) Sparking at the brushes in a DC machine is mainly due to:
(i) Armature reaction (ii) Short circuit
(iii) Low flux (iv) High voltage

(i) Air gap flux density distribution in an ideal machine is:
(i) Square wave (ii) Triangular
(iii) Sawtooth (iv) Sinusoidal

(j) Lap winding is preferred for:
(i) Low voltage, high current (ii) Low voltage, low current
(iii) High voltage, high current (iv) High voltage, low current.

Q.2 (a) Differentiate statically and dynamically induced EMF. Explain the concept of statically and dynamically induced EMF with suitable diagram. [7]

(b) Derive the expression for the field energy and the co-energy of single excited magnetic system. [7]

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Q.3 (a) Differentiate between a 1-Φ transformer and an autotransformer. How is copper saving done in an autotransformer? What are additive and subtractive polarities in an autotransformer? [7]

(b) A 200 kVA has an efficiency of 98% at full load. If the maximum efficiency occurs at three quarters of full load, Calculate the efficiency at half load. Assume negligible magnetizing current and power factor of 0.8 at all loads. [7]

Q.4 (a) Why is back-to-back test performed on a transformer? Write the steps to conduct the test. In back-to-back test, the wattmeter readings for primary and secondary are 25W and 50W, respectively. Calculate the copper loss of a transformer at half the rated load, if both the transformers used are identical. [7]

(b) With the help of circuit diagram, explain the procedures of open circuit tests and short circuit tests conducted on a 1-Φ transformer. [7]

Following test results are obtained from a 10 kVA, 2500/250, 1- Φ transformer. Calculate the efficiency at the load at 0.8 p.f. :

OC Test: 250 V, 0.8 A, 50 W

SC Test: 60V, 3 A, 45 W

Q.5 (a) Draw and explain the magnetization and load characteristics of separately excited DC machine and self-excited DC machine. [7]

(b) A 200V DC shunt motor has an armature resistance of 0.25Ω and runs at 1100 R.P.M, drawing an armature current of 40A. It is required to reduce the speed to 750 R.P.M. If the armature current remains same, find the additional resistance to be added in series with the armature circuit. [7] 4.75

Q.6 (a) What are the advantages of a bank of three 1-phase transformers over a unit transformer? Discuss the different types of connections employed in 3-phase transformers? [7]

(b) A 5 kVA distribution transformer has a full load efficiency of 90% at which copper loss equals Iron loss. The transformer is loaded 24 hours as mentioned below: [7]

No load for 9 hours

25% of full load for 6 hours

50% of full load for 6 hours

Full load for 3 hours

Calculate all day efficiency of the transformer.

Q.7 (a) Derive the EMF equation and Torque equation of DC Machine. [7]

(b) Discuss the Hopkinson's test of DC motor. Why it is needed? What are its advantages? [7]

Q.8 (a) Discuss the various speed control methods of DC motor. [7]

(b) A DC series motor operates at 800 R.P.M with a line current of 100A from 230V supply. Its armature circuit resistance is 0.15Ω and its field resistance is 0.1Ω. Find the speed at which the motor runs at a line current of 25A, assuming that the flux at this current is 45 per cent of the flux at 100A. [7]

Q.9 Write short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Hysteresis and Eddy current losses

(b) Scott Connection

(c) Armature Reaction in DC Motor.

(d) Parallel operation of Single-phase transformer.

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